

ETE 403

VLSI Design

Reference Notes

References: Brackenbury — VLSI Design · Pucknell & Eshraghian — Basic VLSI Design, 3rd Ed. · Mead & Conway — Introduction to VLSI Systems

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Class 04

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Quick Reference: Voltages, Carriers & Current Flow

This page summarises the roles of key voltages and the direction of carrier and current flow in NMOS and PMOS. Refer back here whenever the chapter notation becomes confusing.

Terminal Voltages — What Each One Does

V (Gate-to-Source)	Control voltage. Creates and controls the inversion channel. Determines whether the transistor is ON or OFF. Does not cause current by itself.
V (Drain-to-Source)	Driving voltage. Provides the lateral electric field that pushes carriers through the channel and produces drain current I. No current flows if $V = 0$, even when the channel is open.
V (Threshold Voltage)	Minimum V needed to create strong inversion and turn the transistor ON. Below V the device is OFF regardless of V.
V (Body-to-Source)	Controls depletion width under the gate oxide. Increasing V raises V (Body Effect). Normally tied to 0V for NMOS and to V for PMOS.

Drain vs. Source — Which Has More?

Higher voltage	Drain — always at higher potential than Source in normal operation.
Wider depletion region	Drain side — more reverse bias between drain n and P-body.
Supplies carriers	Source — the terminal that injects majority carriers into the channel.
Collects carriers	Drain — the terminal that collects carriers exiting the channel.
Conventional current	Flows from Drain Source (NMOS). Opposite to electron movement.

NMOS vs. PMOS — Carrier & Current Flow

Property	NMOS	PMOS
Substrate	P-type	N-type
Majority carriers	Electrons	Holes
Channel type	N-type inversion	P-type inversion
Turn ON condition	$V \geq +V$	$V \leq -V$ (negative gate)
Source location	Lower voltage terminal (Ground side)	Higher voltage terminal (V side)
Electron flow	Source Drain (left to right)	Drain Source (right to left, holes)

Conventional current	Drain Source	Source Drain
Body connected to	Ground (0V)	V (highest supply)

Chapter 2: MOSFET Fabrication

2.1 Silicon Wafer Preparation

MOSFET fabrication begins with pure silicon. Raw silicon is heated inside a furnace until it becomes ultra-pure molten silicon.

To form a perfect crystal structure, a small single-crystal silicon piece called a seed crystal is dipped into the molten silicon. In the Czochralski method, the seed crystal is slowly rotated and pulled upward. As the silicon cools, it copies the crystal structure of the seed and forms a large cylindrical single-crystal ingot called a boule.

The cooled boule is then sliced into very thin circular discs known as silicon wafers. These wafers are polished until their surfaces become extremely smooth.

The silicon wafer acts as the base for fabricating millions of MOSFETs and logic gates. For easier understanding, fabrication steps are usually shown using a cross-sectional side view of the wafer, where different material layers are stacked on the silicon substrate.

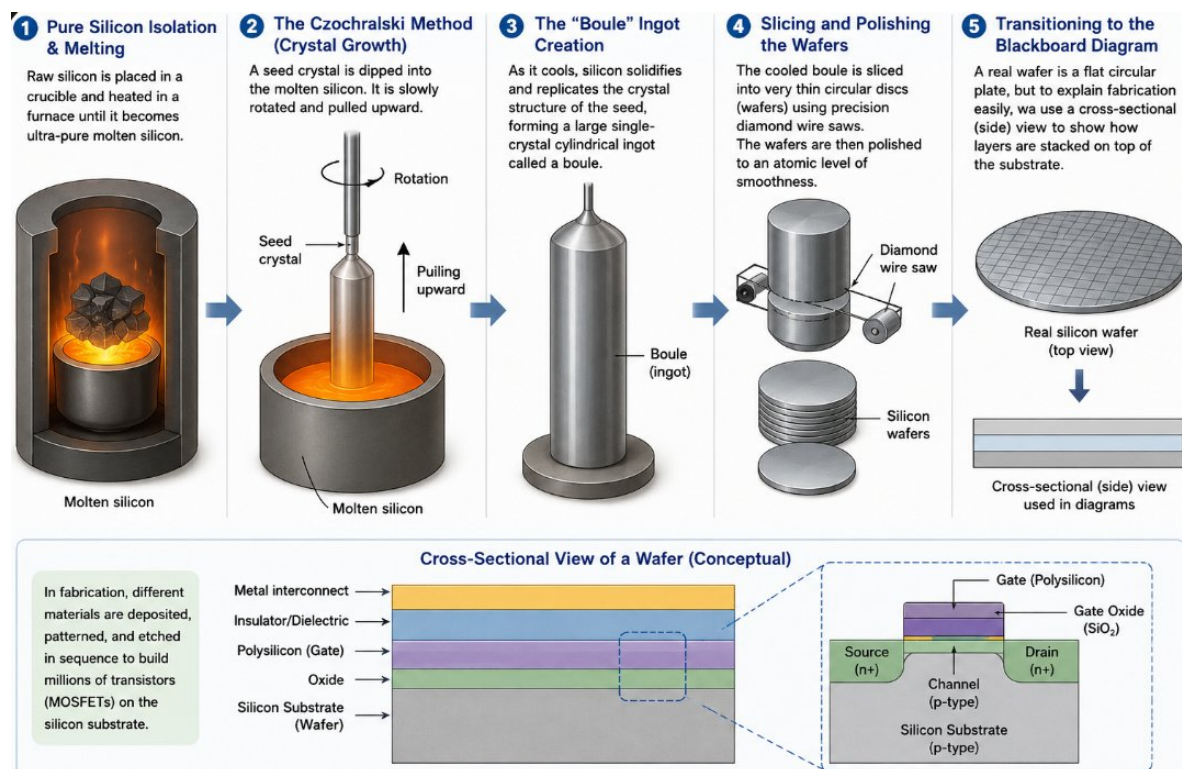


Figure 2.1: MOSFET Fabrication Process & Cross-Sectional View of a Wafer

2.1.1 Wafer Fabrication Process Steps

1 Pure Silicon Isolation & Melting

Raw silicon is placed in a crucible and heated in a furnace until it becomes ultra-pure molten silicon.

2 The Czochralski Method (Crystal Growth)

A seed crystal is dipped into the molten silicon. It is slowly rotated and pulled upward.

3	The “Boule” Ingot Creation As it cools, silicon solidifies and replicates the crystal structure of the seed, forming a large single-crystal cylindrical ingot called a boule.
4	Slicing and Polishing the Wafers The cooled boule is sliced into very thin circular discs (wafers) using precision diamond wire saws. The wafers are then polished to an atomic level of smoothness.
5	Transitioning to the Blackboard Diagram A real wafer is a flat circular plate, but to explain fabrication easily, we use a cross-sectional (side) view to show how layers are stacked on top of the substrate.

2.1.2 Cross-Sectional Layers of a Wafer

Metal Interconnect	Topmost conducting layer that connects different components on the chip.
Insulator / Dielectric	Electrically isolating layer deposited between conducting layers.
Polysilicon (Gate)	Conductive gate material deposited above the gate oxide.
Oxide	Thin insulating oxide layer; acts as gate dielectric (SiO).
Silicon Substrate	The base wafer (p-type or n-type silicon) on which all layers are built.

2.2 MOSFET Fabrication Steps

MOSFET fabrication begins with a P-type silicon substrate, which acts as the body or bulk of the transistor. This substrate is formed by doping silicon with trivalent impurities, making holes the majority carriers.

A very thin layer of Silicon Dioxide (SiO_2) is then grown on top of the substrate. This layer acts as an insulator and prevents direct current flow between the gate and the substrate.

Next, a conductive polysilicon layer is deposited over the oxide layer to form the gate terminal. Although MOS stands for Metal-Oxide-Semiconductor, modern MOSFETs usually use polysilicon instead of metal for the gate.

After this, heavily doped n regions are created on both sides of the gate by diffusing pentavalent impurities into the P-substrate. These n regions form the Source and Drain terminals. The gate structure acts as a self-aligning mask to keep the source and drain properly separated.

Finally, small windows are etched through the oxide layer to expose the gate, source, and drain regions. Metal contacts such as aluminum or copper are then added to provide external electrical connections.

From top to bottom, the MOSFET structure consists of metal contacts, polysilicon gate, silicon dioxide (SiO_2) insulating layer, n source/drain regions, and the P-type substrate.

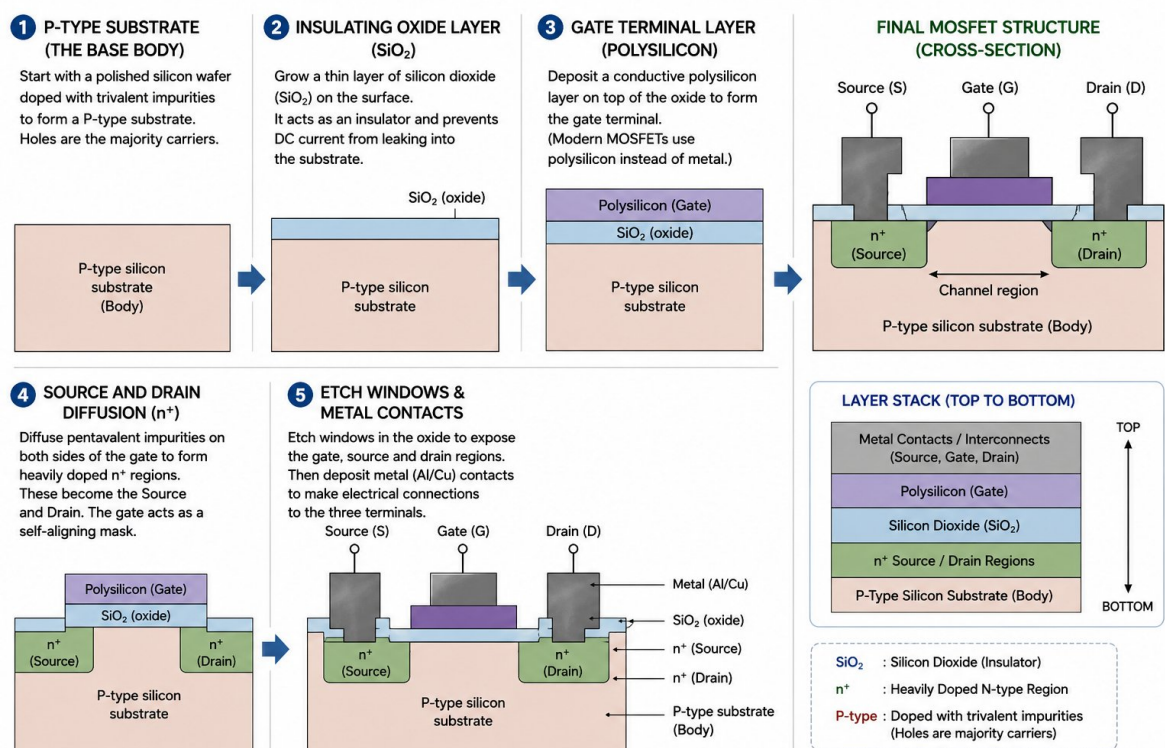


Figure 2.2: MOSFET Fabrication Steps & Layer Stack

2.3 Source and Drain Identification

2.3.1 Symmetric Equality

The Source and Drain regions of a MOSFET are physically identical inside the silicon structure. Their names are decided only by the direction of majority carrier flow.

The Source is the terminal that supplies majority carriers to the channel, while the Drain collects them.

2.3.2 NMOS Rule (Majority Carriers = Electrons)

In an NMOS transistor, electrons are the majority carriers. Since electrons flow from lower voltage to higher voltage, the terminal connected to Ground (0V) acts as the Source, and the higher voltage terminal acts as the Drain.

2.3.3 PMOS Rule (Majority Carriers = Holes)

In a PMOS transistor, holes are the majority carriers. Since holes flow from higher voltage to lower voltage, the terminal connected to the higher supply voltage (V) acts as the Source.

2.4 Enhancement-Mode NMOS: Channel Formation

Enhancement-mode NMOS is normally OFF when no gate voltage is applied ($V_{GS} = 0V$). This happens because there is no conductive path between the two n regions (Source and Drain), and the depletion regions block current flow.

The Gate, oxide layer (SiO_2), and P-substrate together behave like a parallel-plate capacitor.

When a positive Gate-to-Source voltage ($+V_{GS}$) is applied, positive charges accumulate on the gate and create an electric field.

This electric field pushes the majority holes in the P-substrate downward, increasing the depletion region near the oxide surface. At the same time, minority electrons are attracted upward toward the oxide layer from the substrate and the n regions.

As more electrons gather beneath the oxide layer, a thin conductive N-type channel is formed between the Source and Drain. This channel provides a path for current flow through the transistor.

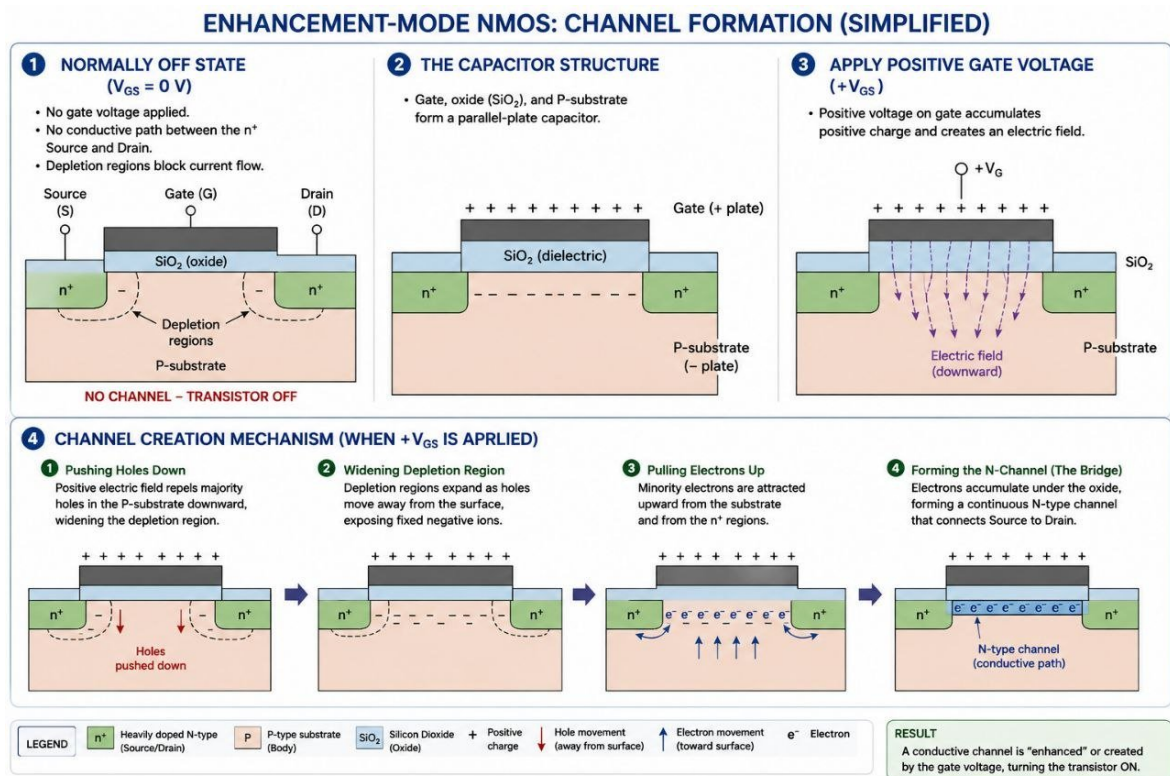


Figure 2.3: Enhancement-Mode NMOS Channel Formation (Simplified)

2.5 Population Inversion & Threshold Voltage

A positive gate voltage attracts minority electrons toward the oxide surface of the P-type substrate. At first, these electrons neutralize the majority holes, making the surface electrically neutral (intrinsic). With further increase in V_{GS} , more electrons accumulate and outnumber holes. The surface then temporarily behaves like an N-type region. This effect is called Population Inversion.

The minimum gate voltage needed to create this strong inversion layer is called the Threshold Voltage (V_{TH}). When $V_{GS} \geq V_{TH}$, a conductive N-channel forms between the Source and Drain.

Population Inversion is also a common term in Laser Diodes, where electrons move to higher energy states for light amplification.

2.5.1 MOS Capacitor Action

When V_{GS} is applied, the gate electrode builds a positive charge plate. The resulting electric field pushes majority holes in the P-substrate downward, clearing the surface and exposing uncompensated static negative acceptor ions. The gate, oxide, and substrate together behave as a parallel-plate capacitor storing this charge configuration.

2.5.2 Weak vs. Strong Inversion

Weak Inversion ($V < V_{TH}$)	The surface becomes less P-type than the bulk. Hole concentration decreases but electrons do not yet dominate. No conductive channel forms.
Strong Inversion ($V \geq V_{TH}$)	Minority electrons from the substrate and the flanking n regions rush to the oxide surface, outnumbering baseline holes. The surface inverts into a temporary N-type bridge — the inversion channel.

2.5.3 The Operational Split: V_{GS} vs. V_{DS}

V_{GS} is solely responsible for creating the inversion channel. However, current will only flow if a lateral driving force V_{DS} is applied to push those electrons across from Source to Drain. Without V_{DS} , the channel exists but carries no current.

V_{GS}	Creates and controls the inversion channel. Determines whether the device is ON or OFF.
V_{DS}	Provides the lateral electric force that drives electrons through the channel. Determines current magnitude.

2.5.4 Summary: Channel Inversion Mechanics

When a positive V_{GS} is applied, the gate creates a positive electric field. This field pushes holes in the P-substrate downward, exposing fixed negative acceptor ions near the oxide surface.

If V_{GS} is small, the surface only becomes less P-type. But when V_{GS} exceeds the Threshold Voltage (V_{TH}), strong inversion occurs. Minority electrons are attracted toward the oxide surface from the substrate and the n regions.

As these electrons outnumber the holes, the surface temporarily behaves like an N-type region, forming a conductive channel between the Source and Drain.

V_{GS} is responsible for creating the channel, while V_{DS} provides the driving force that moves electrons through the channel and produces current flow.

Channel Inversion Mechanics in NMOS Transistor

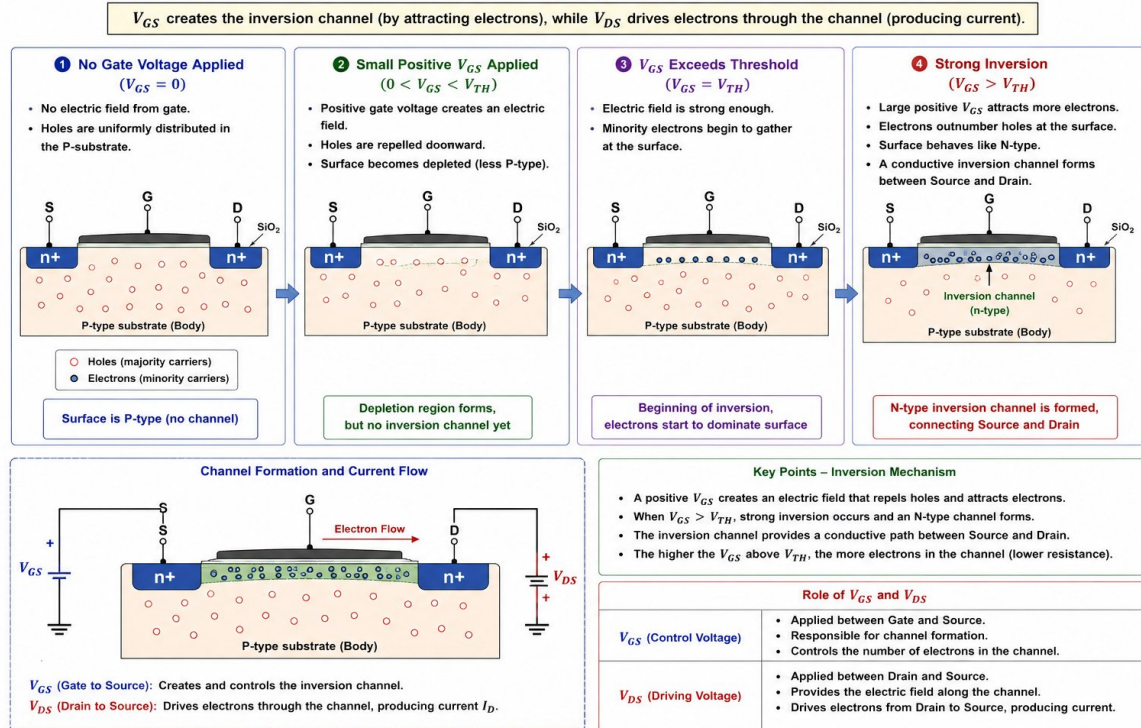


Figure 2.5: Channel Inversion Mechanics in NMOS Transistor

MOSFET: POPULATION INVERSION & THRESHOLD VOLTAGE (SIMPLIFIED)

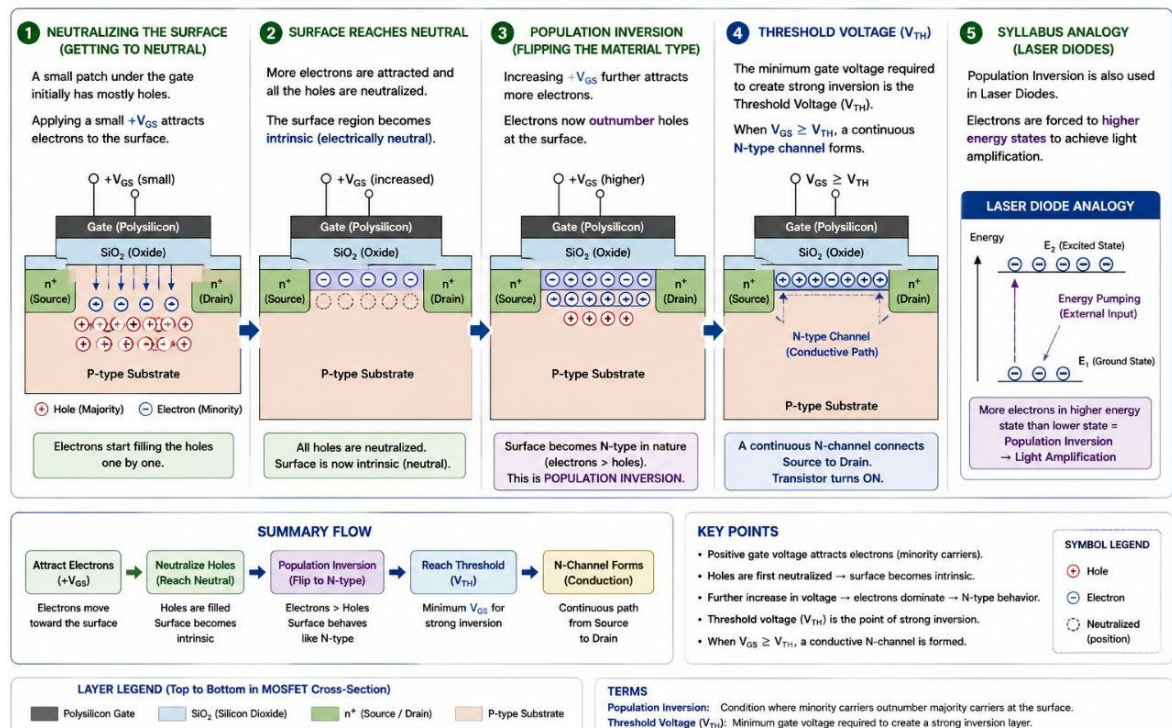


Figure 2.4: MOSFET Population Inversion & Threshold Voltage (Simplified)

2.6 The Fourth Terminal: Substrate (Body/Bulk)

A MOSFET is a 4-terminal device consisting of the Source, Gate, Drain, and Substrate (Body/Bulk).

If the substrate is left floating, it can collect unwanted noise, static charge, or interference from nearby signals. This makes the body voltage unstable and unpredictable.

In an NMOS transistor, the P-type body and N-type source form an internal P-N junction. If the floating body becomes slightly positive compared to the source, this junction becomes forward-biased and turns ON.

As a result, a large unwanted leakage current flows through the chip, causing overheating, malfunction, and possible permanent damage.

To prevent this problem, the substrate body is always connected to a fixed safe voltage so the internal junctions remain reverse-biased:

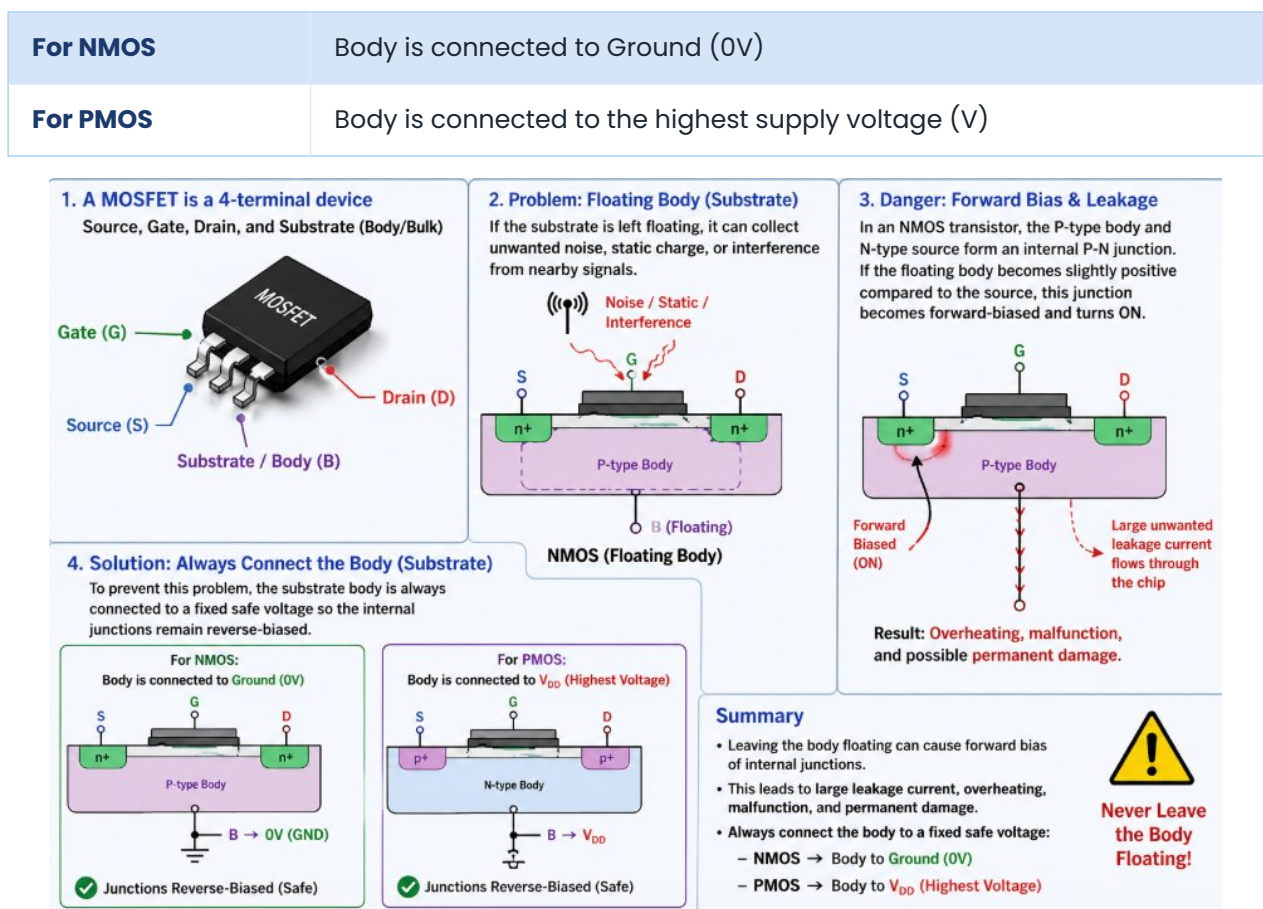


Figure 2.5: MOSFET Body Connection — Floating Body Problem & Solution

2.7 Effect of Drain Voltage on Channel Shape

When no drain voltage is applied ($V_{DS} = 0V$), the electric field created by the gate is uniform across the entire channel region. As a result, a symmetrical rectangular channel forms between the Source and Drain.

When a positive drain voltage ($V_{DS} > 0$) is applied, a horizontal voltage drop develops along the channel. The channel voltage gradually increases from 0V at the Source side to V_{DS} at the Drain side.

At any point in the channel, the effective voltage controlling the channel is:

$$V_{GS} - V(x)$$

Near the Source, this value is high, so the channel remains wide and strong. Near the Drain, the channel voltage becomes larger, reducing the effective gate control to:

$$V_{GS} - V_{DS}$$

Because the electric field becomes weaker near the Drain, the channel narrows and tapers down at the drain side, creating an asymmetric channel shape.

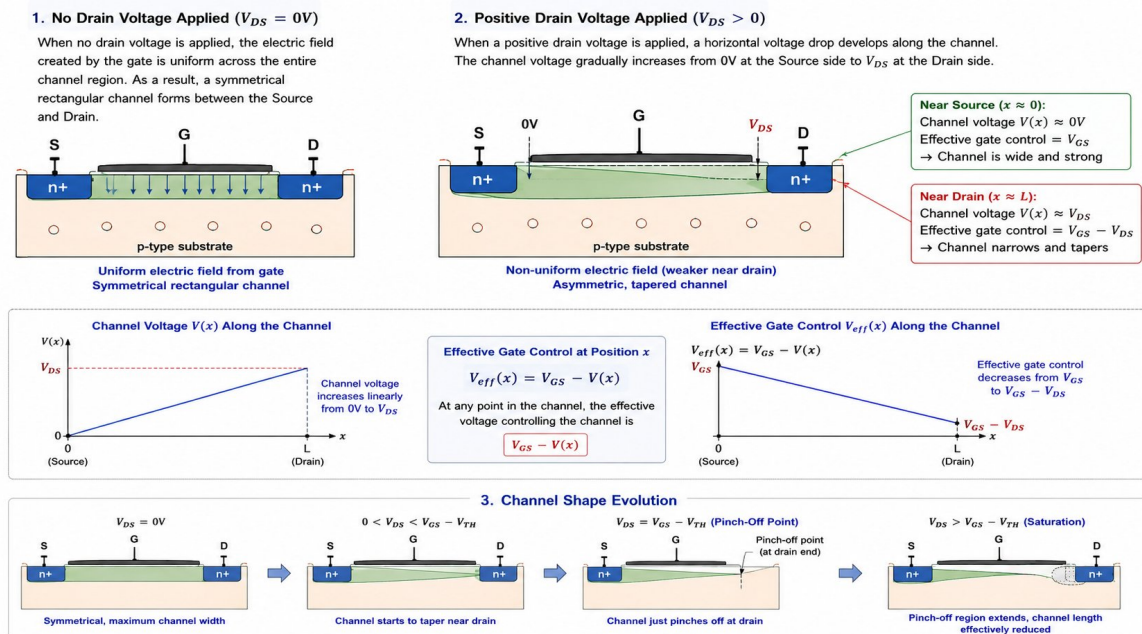


Figure 2.6: Effect of V on Channel Shape – Uniform, Tapered, Pinch-Off & Saturation

2.8 MOSFET Operating Regions & Output Characteristics

The output characteristics of a MOSFET show the relationship between Drain Current (I_D) and Drain-to-Source Voltage (V_{DS}) for different values of Gate-to-Source Voltage (V_{GS}).

Increasing V_{GS} increases the number of inversion electrons in the channel. Higher V_{GS} creates a stronger channel and allows larger drain current (I_D). Low V_{GS} produces a weak channel, causing pinch-off at lower V_{DS} . High V_{GS} lowers channel resistance and shifts saturation further upward.

2.8.1 Cutoff Region (Device OFF)

Condition:

$$V_{GS} < V_{TH}$$

The gate voltage is too small to create population inversion. No conductive channel forms between Source and Drain, so the MOSFET behaves like an open circuit.

$$I_D = 0$$

Example — Case A:

Given	$V = 0.5V, V = 1V$
Check	$0.5V < 1V \quad V < V$
Result	Device OFF, no channel, $I = 0A$

2.8.2 Linear (Triode) Region

Condition:

$$V_{GS} \geq V_{TH} \text{ and } V_{DS} < V_{GS} - V_{TH}$$

A continuous inversion channel forms between Source and Drain. The MOSFET behaves like a voltage-controlled resistor. Increasing V_{DS} increases I_D .

Example — Case D:

Given	$V = 2V, V = 0.5V$
Gate Overdrive	$V - V = 2V - 1V = 1V$
Check	$0.5V < 1V$ Linear Region
Result	Channel fully continuous

Example — Case E:

Given	$V = 5V, V = 2V$
Gate Overdrive	$V - V = 5V - 1V = 4V$
Check	$2V < 4V$ Linear Region

Result	Strong inversion, low-resistance channel, higher drain current
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2.8.3 Pinch-Off Point (Boundary Condition)

Condition:

$$V_{DS} = V_{GS} - V_{TH}$$

At the drain edge, the channel thickness becomes zero. This point forms the boundary between the Linear and Saturation regions.

Example — Case B:

Given	$V = 2V, V = 1V$
Gate Overdrive	$V - V = 2V - 1V = 1V$
Check	$1V = 1V$ Exactly at Pinch-Off
Result	Channel just collapses at the drain edge

2.8.4 Saturation Region

Condition:

$$V_{GS} \geq V_{TH} \text{ and } V_{DS} > V_{GS} - V_{TH}$$

The drain voltage exceeds the channel's maximum support voltage. The pinch-off point moves inward toward the source. The voltage drop across the active channel remains fixed at $V_{GS} - V_{TH}$. Any additional drain voltage appears across the depletion region near the drain. As a result, the drain current becomes nearly constant:

$$I_D \propto (V_{GS} - V_{TH})^2$$

Example — Case C:

Given	$V = 2V, V = 5V$
Gate Overdrive	$V - V = 2V - 1V = 1V$
Check	$5V > 1V$ Saturation Region
Result	Pinch-off occurs; channel voltage fixed at 1V; remaining 4V drops across depletion region

2.8.5 Final Review Table

Mapped Case Study	Inputs Provided	Calculated Gate Overdrive ($V_{GS} - V_{TH}$)	Core Mathematical Condition Verified	Resulting Operational Region	Physical Channel Profile Status
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Case A	$V_{GS} = 0.5V$ $V_{DS} = 1V$	N/A (Device is OFF)	$V_{GS} < V_{TH}$ ($0.5V < 1V$)	Cutoff ($I_D = 0$)	No population inversion; channel cannot form.
Case D	$V_{GS} = 2V$ $V_{DS} = 0.5V$	$2V - 1V = 1V$	$V_{DS} < V_{GS} - V_{TH}$ ($0.5V < 1V$)	Linear (Triode)	Completely open and continuous from source to drain.
Case E	$V_{GS} = 5V$ $V_{DS} = 2V$	$5V - 1V = 4V$	$V_{DS} < V_{GS} - V_{TH}$ ($2V < 4V$)	Linear (Low Resistance)	Highly inverted, deep, continuous channel structure.
Case B	$V_{GS} = 2V$ $V_{DS} = 1V$	$2V - 1V = 1V$	$V_{DS} = V_{GS} - V_{TH}$ ($1V = 1V$)	Pinch-Off Boundary	Inversion layer thickness hits zero precisely at drain edge.
Case C	$V_{GS} = 2V$ $V_{DS} = 5V$	$2V - 1V = 1V$	$V_{DS} > V_{GS} - V_{TH}$ ($5V > 1V$)	Saturation (I_D Constant)	Zero-point detaches and shifts inward; gap near drain.

MOSFET Operating Regions – Complete Analysis with Lecture Cases

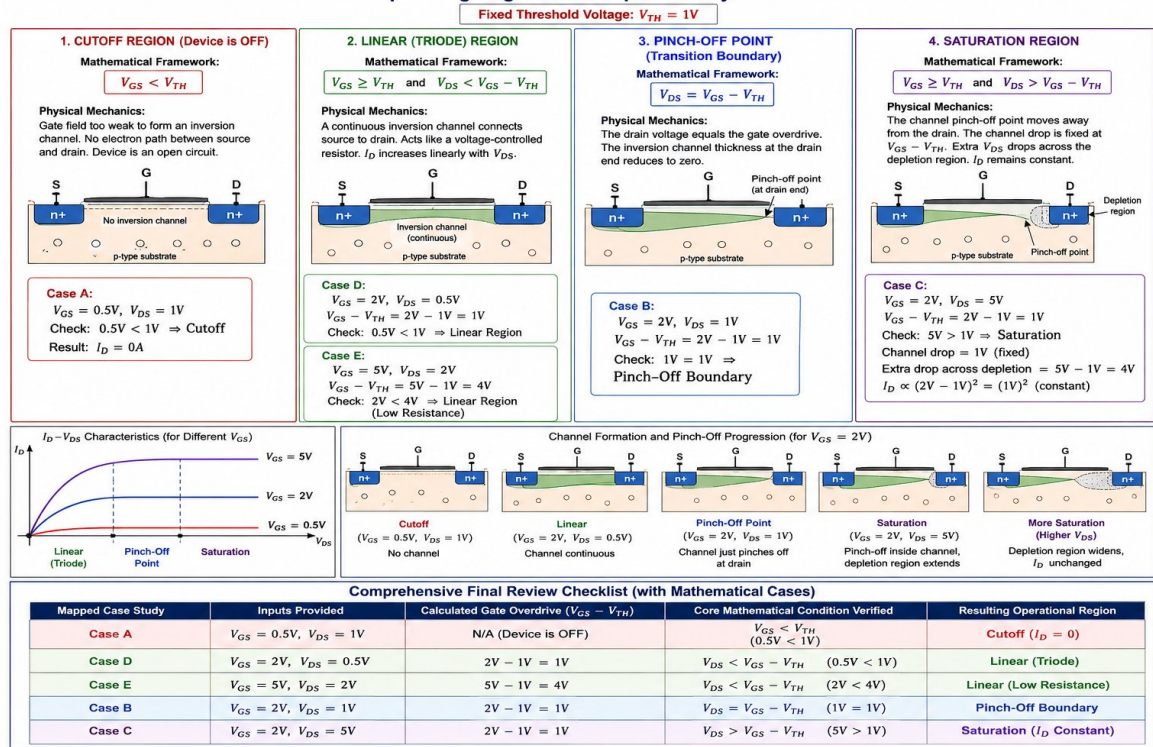


Figure 2.7: MOSFET Operating Regions – Complete Analysis with Lecture Cases

2.9 Water Tap Analogy (V_{GS} vs. V_{DS})

To understand the roles of V_{GS} and V_{DS} , a MOSFET can be compared to a water tap system.

2.9.1 Tap Handle (V_{GS})

The Gate-to-Source voltage (V_{GS}) acts like the tap handle. Its job is to create and control the channel inside the MOSFET.

$V < V$	The channel remains closed, just like a shut tap. No current flows even if drain voltage is present.
V increases	Widens the channel and lowers channel resistance.

2.9.2 Water Pressure (V_{DS})

The Drain-to-Source voltage (V_{DS}) acts like water pressure in the supply tank. It is the actual driving force that pushes electrons through the channel. Even if the channel is fully open, no current flows when:

$$V_{DS} = 0V$$

2.9.3 Summary

2.10 The Gate Material: From Metal to Polysilicon

In the original MOS structure, the gate terminal was made of metal, which is why MOS stands for Metal-Oxide-Semiconductor. In modern VLSI fabrication, the metal gate is replaced by a highly conductive material called Polycrystalline Silicon (Polysilicon or Poly).

2.10.1 Why Single-Crystal Silicon is Not Used for the Gate

Creating single-crystal silicon requires extremely high melting temperatures. If silicon were melted directly on top of the thin gate oxide, the heat would damage or melt the underlying substrate and the already-formed Source and Drain regions.

2.10.2 Why Polysilicon is Used

To avoid this problem, the gate material is deposited at a much lower temperature using chemical vapor deposition. Since the silicon cools without enough energy to form one continuous crystal lattice, it naturally forms many small crystal grains instead of one perfect crystal. This structure is called Polycrystalline Silicon.

2.11 The Insulating Oxide (SiO) & High Input Impedance

A thin layer of highly pure Silicon Dioxide (SiO) is placed between the gate and the semiconductor substrate. This oxide layer acts as a perfect electrical insulator.

Because of this insulating barrier, no direct DC current flows from the gate into the channel. As a result, MOSFETs have extremely high input impedance and require very little input power.

The gate, oxide, and substrate together form a capacitor. When charge is placed on the gate, the oxide prevents it from leaking away quickly. This ability to store charge makes DRAM (Dynamic Random Access Memory) and dynamic logic circuits possible.

In contrast, a BJT requires continuous base current for operation. This constant current flow quickly discharges stored charge, making dynamic memory operation difficult.

2.12 The CMOS Inverter as the Foundational Building Block

The study of MOSFET device physics establishes the behavior of individual transistors in isolation. The next logical step is to examine how these transistors are combined to construct functional digital circuits. The CMOS Inverter, or NOT gate, serves as the primary vehicle for this study.

2.12.1 The Inverter as the Elementary Logic Unit

The CMOS Inverter is the most elementary circuit in digital logic design. A thorough understanding of its operation, layout, and mathematical behavior is considered foundational for two principal reasons:

1	Blueprint for Complex Gates The inverter is operationally identical in principle to more complex gates such as NAND and NOR. No fundamentally new transistor physics is introduced when scaling to multi-input gates. Layout rules, switching conditions, and current behavior remain the same; only the number of transistors and their series or parallel arrangement differ.
2	Complete Scope of Analysis The inverter serves as the test circuit for three core engineering competencies: (1) Physical construction on a silicon substrate, (2) Dynamic performance evaluation — propagation delay t_p , (3) Mathematical derivations — noise margins and switching threshold V_{th} .

2.12.2 Universality of MOSFET Physics Across CMOS Gates

All standard CMOS logic gates are constructed from the same PMOS and NMOS transistor pair topology. The transition from an inverter to a NAND or NOR gate requires no new physical principles — transistors are simply connected in different series or parallel configurations.

CMOS Inverter	1 PMOS + 1 NMOS
CMOS NAND	2 PMOS (parallel) + 2 NMOS (series)
CMOS NOR	2 PMOS (series) + 2 NMOS (parallel)

2.13 Pure Silicon & N-Type Doping

Pure silicon is a poor conductor at room temperature because most electrons are tightly bound in covalent bonds. Only a small number of free electrons and holes are produced naturally by thermal energy.

To increase conductivity, silicon is doped with pentavalent impurity atoms. These atoms have five valence electrons. Four electrons form covalent bonds with silicon atoms, while the fifth electron remains free and can move easily, increasing conductivity. This forms N-type semiconductor material.

Even though N-type material contains many free electrons, it is electrically neutral overall because each donor atom also contains a balancing positive proton charge.

Holes are still present in N-type material due to normal thermal generation, but they exist only as minority carriers.

2.14 Structural Layout of the NMOS Transistor

An NMOS transistor is built on a lightly doped P-type substrate, also called the Body or Bulk. Two heavily doped n regions are formed on opposite sides of the substrate. These regions act as the Source and Drain terminals.

The Drain is considered the output terminal of the device, while the Source acts as the common reference terminal for both input and output operations.

The control voltage is applied between the Gate and Source:

$$V_{GS} = V_{\text{Gate}} - V_{\text{Source}}$$

The output driving voltage is applied between the Drain and Source:

$$V_{DS} = V_{\text{Drain}} - V_{\text{Source}}$$

The gate voltage controls channel formation, while the drain voltage drives current through the channel.

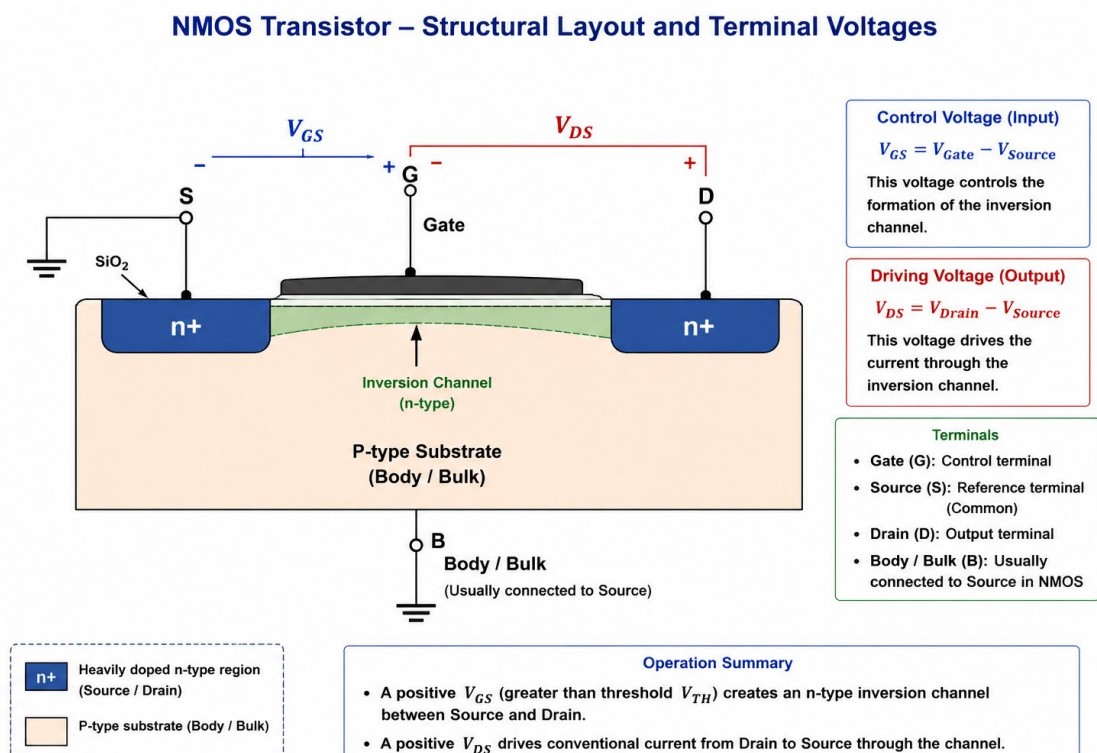


Figure 2.8: NMOS Transistor – Structural Layout and Terminal Voltages

2.15 Depletion Isolation & Body Effect (V_{BS})

Reverse-biased P-N junctions inside a MOSFET naturally form depletion regions around the Source-Body and Drain-Body interfaces. These regions remove mobile charge carriers, blocking unwanted current and allowing millions of transistors to operate independently on the same chip.

Because the Drain is typically at a higher positive voltage (V_{DS}), the drain-body junction is more reverse-biased than the source-body junction. Consequently, the drain-side depletion region is wider and deeper.

2.15.1 Body Effect

The Body Effect is the change in Threshold Voltage (V_{TH}) caused by a change in the body-to-source voltage (V_{BS}). When V_{BS} increases, the depletion region under the gate oxide widens. A wider depletion region requires more gate voltage to create inversion, raising V_{TH} .

$$V_{TH} = V_{TH0} + \gamma (\sqrt{|2\phi_F + V_{BS}|} - \sqrt{|2\phi_F|})$$

V_{TH0}	Threshold voltage when $V_{BS} = 0$
γ	Body effect coefficient
ϕ_F	Fermi potential of the substrate

Depletion Isolation & Body Effect (V_{BS}) in NMOS

Reverse-biased depletion regions provide isolation. Changing body-to-source voltage (V_{BS}) changes threshold voltage (V_{TH}).

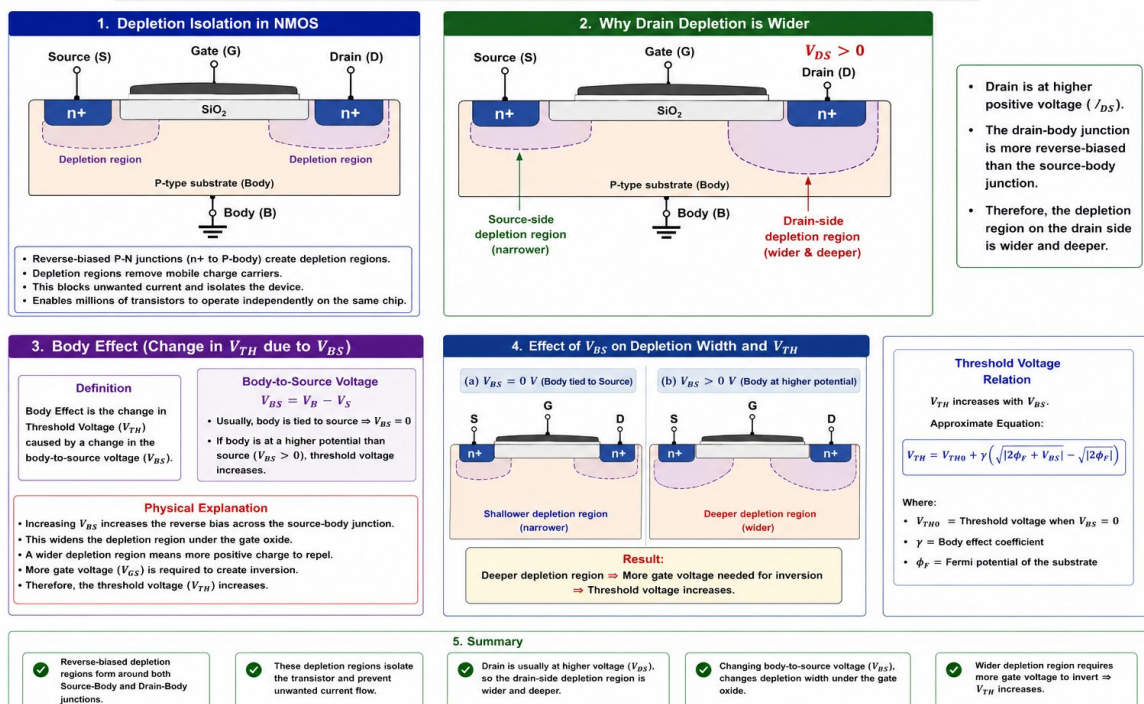


Figure 2.9: Depletion Isolation & Body Effect (V_{BS}) in NMOS

Appendix: Job Written & Viva Questions

Questions likely to appear in written tests and viva boards at semiconductor and VLSI companies such as Ulkasemi Pvt. Limited, Neural Semiconductor Limited, and similar firms, based on the topics covered in this chapter.

A.1 Silicon Fabrication & Wafer Preparation

Q: What is the Czochralski method and why is it used?

A: It is the process of growing a single-crystal silicon ingot by slowly rotating and pulling a seed crystal from molten silicon. It is used because it produces a defect-free, uniform crystal structure essential for reliable transistor fabrication.

Q: Why is polysilicon used as the gate material instead of single-crystal silicon?

A: Single-crystal silicon requires extremely high melting temperatures. Depositing molten silicon on top of the thin gate oxide would destroy the substrate and diffused regions. Polysilicon is deposited at a much lower temperature via chemical vapor deposition, forming many small crystal grains — adequate conductivity without thermal damage.

Q: What are the five material layers of a MOSFET cross-section from top to bottom?

A: Metal contacts / interconnects Polysilicon gate Silicon Dioxide (SiO₂) n Source/Drain regions P-type silicon substrate.

Q: What is the role of the SiO₂ layer?

A: It acts as a perfect insulator between the gate and the substrate, preventing DC gate current. It also forms the dielectric of the gate capacitor, enabling charge storage for DRAM and dynamic logic circuits.

A.2 MOSFET Device Physics

Q: What is threshold voltage V_{th} and what factors affect it?

A: V_{th} is the minimum gate-to-source voltage required to create strong inversion and form a conductive channel. It is affected by oxide thickness, doping concentration, body-to-source voltage (Body Effect), and temperature.

Q: What is the Body Effect?

A: The change in threshold voltage (V_{th}) caused by a non-zero body-to-source voltage (V_{bs}). Increasing V_{bs} widens the depletion region under the gate oxide, requiring more gate voltage to create inversion. Governed by: $V_{th} = V_{th0} + (\sqrt{|2qN_A\epsilon_s|} + |V_{bs}| - \sqrt{|2qN_A\epsilon_s|})$.

Q: Explain Population Inversion in a MOSFET.

A: When V exceeds V_t , minority electrons accumulate at the P-substrate surface faster than holes are present. Electrons outnumber holes, inverting the surface from P-type to N-type behavior. This N-type bridge between Source and Drain is the inversion channel.

Q: What is the difference between V_{gs} and V_{ds} ?

A: V_{gs} is the control voltage that creates and sizes the inversion channel. V_{ds} is the driving voltage that pushes carriers through the channel. A channel can exist ($V_{gs} \geq V_t$) but carry zero current if $V_{ds} = 0$.

Q: Why is the drain-side depletion region wider than the source-side?

A: The drain is at a higher positive voltage ($V_{ds} > 0$), making the drain-body P-N junction more reverse-biased than the source-body junction. A larger reverse bias produces a wider and deeper depletion region on the drain side.

Q: What is channel pinch-off and when does it occur?

A: Pinch-off occurs when $V_{ds} = V_{gs} - V_t$. At this point the effective gate control at the drain edge ($V_{gs} - V_{ds}$) drops to zero, collapsing the channel thickness to zero at the drain. It marks the boundary between the Linear and Saturation regions.

A.3 Operating Regions

Q: Name the three operating regions of a MOSFET and their conditions.

A: Cutoff: $V_{gs} < V_t$ — no channel, $I_D = 0$. Linear (Triode): $V_{gs} \geq V_t$ and $V_{ds} < V_{gs} - V_t$ — channel open, I_D rises with V_{ds} . Saturation: $V_{gs} \geq V_t$ and $V_{ds} > V_{gs} - V_t$ — channel pinched off, $I_D \propto (V_{gs} - V_t)^2$.

Q: In saturation, why does drain current remain nearly constant despite increasing V_{ds} ?

A: Once pinch-off occurs, the voltage across the active channel portion is fixed at $V_{gs} - V_t$. Any additional V_{ds} drops across the depletion region near the drain and does not increase the channel voltage. Since channel voltage is fixed, the carrier velocity and hence current remain essentially constant.

Q: A MOSFET has $V_t = 1V$, $V_{gs} = 3V$, $V_{ds} = 1.5V$. Which region is it in?

A: Gate overdrive = $3 - 1 = 2V$. Since $V_{ds} (1.5V) < \text{overdrive} (2V)$, the device is in the Linear (Triode) region.

Q: What happens to V_t if the body voltage is raised above the source voltage in NMOS?

A: V_t increases due to the Body Effect. The wider depletion region under the gate oxide demands more gate voltage to achieve inversion.

A.4 CMOS & Circuit Design

Q: Why is the CMOS Inverter considered the foundational circuit in VLSI?

A: All CMOS logic gates (NAND, NOR, complex gates) use the same MOSFET physics, same layout rules, and same switching conditions as the inverter. Moving from an inverter to a NAND or NOR gate only changes the number of transistors and their series/parallel arrangement — no new device physics is introduced.

Q: How does a CMOS Inverter differ from a CMOS NAND gate structurally?

A: A CMOS Inverter uses 1 PMOS + 1 NMOS. A CMOS NAND gate uses 2 PMOS in parallel + 2 NMOS in series. The NAND adds one transistor per type but the same complementary switching principle applies.

Q: Why does a MOSFET have extremely high input impedance compared to a BJT?

A: The SiO₂ gate oxide is a perfect insulator, so no DC current flows into the gate terminal. A BJT requires continuous base current for operation, giving it much lower input impedance. The MOSFET's insulated gate also enables charge storage, making DRAM possible.

Q: Why must the body of an NMOS transistor never be left floating?

A: A floating body can accumulate noise or static charge, raising its potential above the source. This forward-biases the internal P-N junction between the P-body and the n source, turning it ON and causing large leakage current, overheating, and potential permanent damage.

Q: In NMOS, which terminal is Source and which is Drain, and how do you determine it?

A: The Source is the terminal connected to the lower voltage (Ground side) because electrons — the majority carriers — are injected from there into the channel. The Drain is at the higher voltage and collects those electrons. The labels are not fixed physically; they depend on applied voltages.

A.5 Viva One-Liners (Quick Fire)

What does MOS stand for?	Metal-Oxide-Semiconductor.
What replaces metal in modern MOSFETs?	Polysilicon (Polycrystalline Silicon).
What are majority carriers in NMOS?	Electrons.
What are majority carriers in PMOS?	Holes.
Direction of conventional current in NMOS?	Drain Source.
Direction of electron flow in NMOS?	Source Drain.
What is a boule?	A large single-crystal silicon ingot grown by the Czochralski method.

What does V control?	Channel formation and width (device ON/OFF).
What does V control?	Current magnitude through the channel.
When is $I \propto (V - V_t)^2$?	In the Saturation region.
What is strong inversion?	The condition where electrons outnumber holes at the oxide surface, forming a conductive N-type channel.
Body connected to Ground in which device?	NMOS.
Body connected to V in which device?	PMOS.